Financial Management

End Term Assignment

1 NPV Calculation and Project Selection

Years	Project A	Project B		Discount Rate	NPV Project A	NPV Project B
0	-2200000	-2700000		15%	₹ 296,251.84	₹212,293.81
1	600000	700000		16%	₹ 223,139.26	₹126,995.81
2	600000	700000		17%	₹ 153,428.08	₹45,666.09
3	600000	700000		18%	₹ 86,916.56	-₹31,930.68
4	600000	700000		19%	₹ 23,417.08	-₹106,013.40
5	600000	700000		20%	-₹ 37,244.94	-₹176,785.77
6	600000	700000				
7	600000	700000				

1.1 Analysis and Recommendation

"Based on the NPV calculations displayed above, **Project A is the preferred investment**.

1. **Higher Returns:** Project A generates a consistently higher Net Present Value (NPV) than Project B at every discount rate tested (15% to 20%).
2. **Risk Profile:** Project B becomes unviable (negative NPV) at a discount rate of 18%, whereas Project A remains profitable (positive NPV) up to 19%. This indicates that Project A offers a better safety margin against rising costs of capital.

Therefore, since Project A maximizes shareholder wealth more effectively and sustains profitability at higher discount rates, it should be selected over Project B."

2 ValuTech Solutions Dividend Policy Analysis

This section analyzes the dividend policy decision for ValuTech Solutions using three major theories: Modigliani-Miller (M&M), Walter's Model, and Gordon's Model.

2.1 Modigliani-Miller (M&M) Theory in a Perfect Market

The Modigliani-Miller dividend irrelevance theory suggests that in a perfect market—where there are no taxes, transaction costs, or information asymmetry—a firm's dividend policy does not affect its value.

According to this theory, the value of the firm is determined solely by its earning power and the risk of its underlying assets, not by how it distributes its earnings (Van Horne & Wachowicz, 2008).

- **Impact on Share Price:** M&M argues that investors can create their own desired cash flow positions (homemade dividends) by selling shares if they need cash or reinvesting dividends if they do not. Therefore, ValuTech's decision to pay dividends or retain earnings to fund the project would have no impact on shareholders' wealth or the share price in a perfect market.
- **Advice on Funding:** M&M theory would support cutting the dividend to fund the new project if the project has a positive Net Present Value (NPV). Since the source of financing (retained earnings vs. new equity) is irrelevant to firm value, management should prioritize taking the profitable investment (ROI 20%) over maintaining a specific dividend payout (Brigham & Ehrhardt, 2013).

2.2 WALTER'S MODEL ANALYSIS

Walter's model emphasizes the relationship between the firm's internal rate of return (r) and its cost of capital (k_e). It suggests that dividend policy is relevant and affects firm value (Van Horne & Wachowicz, 2008).

- **Relationship of r vs. k_e:**
  - ValuTech's project Return on Investment (r) = **20%**.
  - Cost of Capital (k_e) = **15%**.
- **Recommendation:** Since $r > k_e$ (20% > 15%), ValuTech is classified as a "growth firm." Walter's model recommends a **0% dividend payout ratio** (100% retention).
- **Impact on Company Value:** Retaining earnings is optimal because the firm can reinvest funds at a higher rate (20%) than shareholders can obtain elsewhere at the same risk level (15%). By reinvesting all earnings, the firm maximizes shareholder wealth and increases the market price of the share.

2.3 GORDON'S DIVIDEND GROWTH MODEL

Gordon's model also supports the relevance of dividend policy but focuses on the valuation of a growing perpetuity. The formula $P_0 = \frac{D_1}{k_e - g}$ is, where $g = b \times r$ (b is the retention ratio).

- **Optimal Policy:** Similar to Walter's model, for a firm where $r > k_e$, the optimal policy is to retain earnings to drive growth. A higher retention ratio *(b)* leads to a higher growth rate *(g)*, which, when $r > k_e$, results in a higher share price (P_0) (Brigham & Ehrhardt, 2013).
- **Effect of Retention on Growth and Price:**
  - The growth rate g is directly proportional to the retention ratio (g = Retention Ratio × ROI).
  - By increasing the retention ratio to fund the 20% ROI project, ValuTech increases its growth rate. Because the return on these retained earnings (20%) exceeds the discount rate (15%), the value of the expected future growth outweighs the reduction in current dividends, leading to a higher share price.

3 CAPITAL STRUCTURE AND FIRM VALUATION

This question evaluates the value of three identical firms (Alpha, Beta, Delta) with different capital structures using Net Income (NI), Traditional, and Net Operating Income (NOI / M&M without tax) theories.

Given Data:

- **Return on Capital (ROC):** 20%.
- **Alpha Ltd (Unlevered):** 50,000 shares @ Rs. 10 = Rs. 500,000 Equity. Debt = 0.
- **Beta Ltd:** 30,000 shares @ Rs. 10 = Rs. 300,000 Equity. Debt = Rs. 200,000.
- **Delta Ltd:** 25,000 shares @ Rs. 10 = Rs. 250,000 Equity. Debt = Rs. 250,000.
- **Total Capital Employed:** Rs. 500,000 for all firms (implied by identical operations and total face values).
- **EBIT (Earnings Before Interest and Taxes):**
  - EBIT = Total Capital × ROC = 500,000 × 20% = Rs. 100,000.

Cost of Capital Data (k_e and k_d) from Table:

- **Alpha (0% Debt):** k_e = 9%, k_d = 0%.
- **Beta (40% Debt):** Total capital is 500k; Debt is 200k. Debt Ratio = 200,000 / 500,000 = 40%.

- From table at 40%: k_e = 9.75%, k_d = 6.90%.

- **Delta (50% Debt):** Total capital is 500k; Debt is 250k. Debt Ratio = 250,000 / 500,000 = 50%.
  - From table at 50%: k_e= 11%, k_d = 8.50%.

3A. Evaluation of Firm Value

1. Net Income (NI) Approach

The NI approach assumes that increasing debt (cheaper than equity) lowers the overall Weighted Average Cost of Capital (WACC) and increases firm value.

- **Formula:** V = S + D (Value of Equity + Value of Debt).
- $S = \frac{Earnings\ available\ for\ equity}{k_e} = \frac{EBIT - Intere}{k_e}$

Calculations:

- **Alpha Ltd (0% Debt):**
  - Interest = 0.
  - Earnings for Equity = 100,000.
  - S = 100,000 / 0.09 = 1,111,111.
  - **Value (V) = 1,111,111 + 0 = Rs. 1,111,111.**
- **Beta Ltd (40% Debt):**
  - Interest = 200,000 × 6.90% = 13,800.
  - Earnings for Equity = 100,000 - 13,800 = 86,200.
  - S = 86,200 / 0.0975 = 884,103.
  - **Value (V) = 884,103 + 200,000 = Rs. 1,084,103.**
- **Delta Ltd (50% Debt):**
  - Interest = 250,000 × 8.50% = 21,250.
  - Earnings for Equity = 100,000 - 21,250 = 78,750.
  - S = 78,750 / 0.11 = 715,909.
  - **Value (V) = 715,909 + 250,000 = Rs. 965,909.**

2. TRADITIONAL THEORY

The Traditional theory posits that there is an optimal capital structure. Initially, adding debt increases value, but beyond a certain point, the rising cost of equity (k_e) and debt (k_d) due to financial risk decreases total value.

- *Note: The calculations generally follow the same mechanics as the NI approach using the specific k_e and k_d provided in the table for each leverage level.*
- Using the calculations above:
  - **Alpha (0% Debt):** V = Rs. 1,111,111.
  - **Beta (40% Debt):** V = Rs. 1,084,103.
  - **Delta (50% Debt):** V = Rs. 965,909.
- *Observation:* In this specific dataset, the cost of equity rises quickly enough that value decreases as debt is added.

3. MODIGLIANI-MILLER (M&M) WITHOUT TAXES

M&M (No Taxes) states that capital structure is irrelevant. The value of the firm is determined by capitalizing the Operating Income (EBIT) at the overall cost of capital (k_o) appropriate for the firm's business risk class.

- Assume the unlevered firm (Alpha) sets the base capitalization rate (k_o = 9%).
- **Formula:** $V = \frac{EBIT}{k_o}$
- **Calculation:**
  - V = 100,000 / 0.09 = Rs. 1,111,111}.
- **Result:** Under M&M without taxes, Alpha, Beta, and Delta all have the same value of Rs. 1,111,111, regardless of their debt levels.

3B. COMPARISON OF VALUATION ACROSS THEORIES

Does the valuation differ? Yes, the valuation differs significantly across the theories because of their underlying assumptions regarding how investors perceive and price risk.

- **Net Income (NI) Theory:** This theory suggests that value can be increased by using debt because it assumes that the cost of debt (k_d) and cost of equity (k_e) remain constant regardless of leverage

(Van Horne & Wachowicz, 2008). This ignores the increasing financial risk born by shareholders, artificially inflating the firm's value as cheaper debt is added.

- **Traditional Theory:** This approach adopts a middle ground, acknowledging that while debt is cheaper initially, excessive leverage causes the cost of equity (k_e) to rise significantly due to financial risk. According to the Traditional view, there is an optimal capital structure where the Weighted Average Cost of Capital (WACC) is minimized and firm value is maximized (Brigham & Ehrhardt, 2013). In this specific dataset, the "optimal" point appears to be at zero debt (Alpha), as k_e rises quickly enough to diminish value at higher debt levels.
- **M&M (No Tax):** M&M argue that in a friction-less market, capital structure is irrelevant. They state that the increase in expected return on equity from leverage is exactly offset by the increase in financial risk, leaving the total value of the firm constant (Modigliani & Miller, 1958). This explains why, under the M&M approach, Alpha, Beta, and Delta all share the same value (Rs. 1,111,111) regardless of their debt ratios.

Summary: The valuation differs because NI Theory ignores the full impact of financial risk on k_e, **Traditional Theory** accounts for varying costs but allows for an optimal structure, and **M&M Theory** assumes perfect markets where arbitrage forces all firm values to be identical regardless of leverage16.

REFERENCES

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